

TOMMY M^cMICHEN

University of Illinois Urbana-Champaign · Siebel School of Computing and Data Science

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RESEARCH INTERESTS

My interests lie in *compilers*, specifically how we can leverage modern programming languages to design general-purpose *intermediate representations* that (1) improve the precision of *static analysis* by preserving high-level information, (2) enable novel *optimizations* on the layout and organization of memory, and (3) open the door to *compiler-runtime codesigns* that rethink existing hardware abstractions.

PROFESSIONAL EXPERIENCE

University of Illinois Urbana-Champaign, Urbana, Illinois, USA 2026 – Present
Postdoctoral Research Associate

- Automatic Compiler Construction

Meta, Menlo Park, California, USA Summer 2025
Software Engineering Intern, Programming Languages and Runtimes, Android Native Compiler Team

- Improved the representation of ClangIR, an open-source C/C++ MLIR compiler.
- Developed analyses and transformations for C++ move semantics in ClangIR.
- Lead ClangIR open-source development efforts through issue creation and code reviews.

Texas Instruments, Dallas, Texas, USA Summer 2019, Summer 2020
Digital Design Engineering Intern, Embedded Processors, Analytics Team

- Performed integration testing for hardware implementation of cache coherence protocol.
- Developed coverage metrics for cache coherence testing.
- Implemented automatic generation of RTL and TLM from descriptor files.

National Instruments, Austin, Texas, USA Summer 2018
R&D Software Engineering Intern, Digitizers

- Designed and implemented FPGA logic for new function generator feature with LabVIEW.
- Added kernel, driver and API support for new function generator feature.
- Implemented full driver stack support for highly-customisable oscilloscope triggers.
- Communicated with multiple teams to add new .NET API entry points.

EDUCATION

Northwestern University, Evanston, Illinois 2020 – 2026
Ph.D. in Computer Science, *Advised by Simone Campanoni* 2026
M.Sc. in Computer Science, *Advised by Simone Campanoni* 2023

Rose-Hulman Institute of Technology, Terre Haute, Indiana 2016 – 2020
B.Sc. in Computer Engineering and Computer Science 2020

PUBLICATIONS

AI Coding Agents Need Better Compiler Remarks, *CoDAIM 2026*.

Akash Deo, Simone Campanoni, **Tommy McMichen**.

Automatic Data Enumeration for Fast Data Collections, *CGO 2026*.



Tommy McMichen, Simone Campanoni.

Saving Energy with Per-Variable Bitwidth Speculation, *ASPLOS 2025*.



Tommy McMichen, David Dlott, Panitan Wongse-ammat, Nathan Greiner, Hussain Khajanchi, Russ Joseph, Simone Campanoni.

Representing Data Collections in an SSA Form, *CGO 2024*.



Tommy McMichen, Nathan Greiner, Peter Zhong, Federico Sossai, Atmn Patel, Simone Campanoni.

Getting a Handle on Unmanaged Memory, *ASPLOS 2024*.



Nick Wanninger, **Tommy McMichen**, Simone Campanoni, Peter Dinda.

Program State Element Characterization, *CGO 2023*.



Enrico Armenio Deiana, Brian Suchy, Michael Wilkins, Brian Homerding, **Tommy McMichen**, Katarzyna Dunajewski, Peter Dinda, Nikos Hardavellas, Simone Campanoni.

NOELLE Offers Empowering LLVM Extensions, *CGO 2022*.



Angelo Matni, Enrico Armenio Deiana, Yian Su, Lukas Gross, Souradip Ghosh, Sotiris Apostolakis, Ziyang Xu, Zujun Tan, Ishita Chaturvedi, Brian Homerding, **Tommy McMichen**, David I. August, Simone Campanoni.

Fine-Grained Acceleration using Runtime Integrated Custom Execution (RICE), *CASES 2019*.

Leela Pakanati, **Tommy McMichen**, Zachary Estrada.

INVITED TALKS

“Representing Data Collections for Analysis and Transformation”

Languages, Systems, and Data Seminar, *University of California, Santa Cruz*.

October 2025

Computer Architecture Group Meeting, *University of Cambridge*.

March 2024

Tech Talk, *Rose-Hulman Institute of Technology*.

October 2023

Student Seminar Series, *Northwestern University*.

October 2023

Constellation Workshop, *Northwestern University*.

July 2023

“Towards Collection-Oriented Compilation in LLVM”

LLVM Developers’ Meeting.

October 2025

TEACHING EXPERIENCE

Teaching Assistant, *COMP_SCI 322 Compiler Construction*, Prof. Simone Campanoni.

Winter 2022

Resident Tutor, *Computer Science and Computer Engineering Departments*.

Aug. 2019 – May 2020

SERVICE

Artifact Evaluation Committee , International Symposium on Code Generation and Optimization (CGO).	2027
Student Volunteer , Workshop on Advancing Theory, Research, and Practice for Generative AI in University-Level Computing Education.	2026
Artifact Evaluation Committee , International Conference on Compiler Construction (CC).	2026
Board Member , Computer Science PhD Advisory Council, Northwestern University.	2025 – 2026
Board Member , Computer Science Social Initiative, Northwestern University.	2021 – 2026
Member , CS Ph.D. Orientation Planning Committee, Northwestern University.	2022 – 2025
Member , CS Ph.D. Visit Day Planning Committee, Northwestern University.	2022 – 2026
Student Volunteer , International Symposium on Microarchitecture (MICRO).	2022
Chairperson , IEEE, Rose-Hulman Institute of Technology student branch.	2019 – 2020
Corresponding Secretary , Eta Kappa Nu (HKN), Epsilon Eta Chapter.	2019 – 2020
Member , Eta Kappa Nu (HKN), Epsilon Eta Chapter.	2018 – 2020

AWARDS AND FUNDING

PhD Student Leadership Award, <i>Northwestern University – CS Department</i> .	2026
PhD Student Research Award, <i>Northwestern University – CS Department</i> .	2026
LLVM Foundation Student Travel Grant, <i>LLVM Developers’ Meeting</i> .	2025
NSF Student Travel Grant, <i>ASPLOS</i> .	2025
NSF Student Travel Grant, <i>HPCA/PPoPP/CGO</i> .	2024
NSF Student Travel Grant, <i>HPCA/PPoPP/CGO</i> .	2023
IP/ROP Student Travel Award.	2019
NSF Student Travel Grant, <i>ESweek</i> .	2019
IP/ROP Student Project Grant.	2018

RESEARCH ADVISING

Leyla Latifova , B.Sc., <i>Fast Translation Validation for Transformation Pipelines</i> .	2026 – Present
Akash Deo , M.Sc., <i>Designing compiler tools for AI-assisted vectorization</i> .	2025 – 2026 (Apple)
Benjamin Ye , M.Sc., <i>Characterizing differences between LLVM front-ends</i> .	2025 – Present
Benjamin Ye , B.Sc., <i>Automatically generating MEMOIR from Rust</i> .	2024 – 2025